

AI-Based Substation Load MONITORING AND PREDICTION

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The growing electricity demand is increasing substation loading, making accurate load forecasting essential for transformer planning and reliable operation. This article presents an AI-based substation load growth prediction and monitoring system using the ESP32-WROOM-32, PZEM-004T, Firebase, and Python. The system monitors transformer load in real time, stores data in the cloud, generates peak-load reports, and predicts future demand using machine learning.

Substations rely on high-capacity

power transformers to distribute electricity efficiently. Rising transformer loads can cause overheating, reduce equipment life, and lead to failures, making continuous monitoring and predictive analysis essential.

Fig. 1 shows a high-voltage

oil-filled power transformer in an outdoor substation. Insulating oil provides cooling and electrical insulation, while radiator fins dissipate heat. High-voltage bushings provide safe electrical connections for reliable power transmission and distribution.

BILL OF MATERIALS

Component	Specification/Description	Quantity
ESP32-WROOM-32	Wi-Fi-enabled microcontroller module	1
PZEM-004T	AC voltage, current, power, and energy measurement module	1
Step-down transformer	230V AC to 24V AC, 1A	1
USB adaptor cable	USB type-A male to micro-USB female adaptor	1



Fig. 1: High-voltage power transformer in an electrical substation

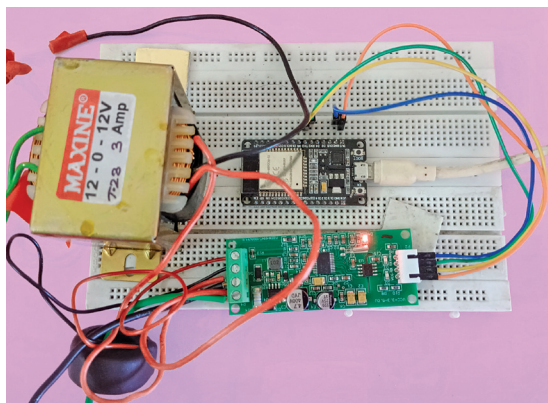


Fig. 2: Author's prototype

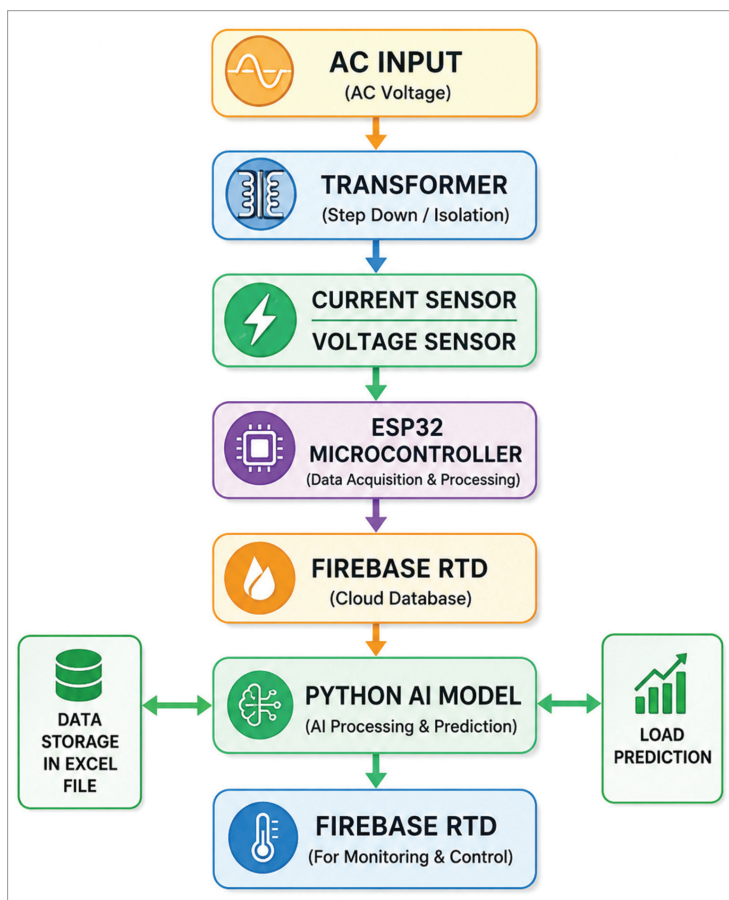


Fig. 3: Block diagram of the proposed system